

CareVoice AI: Voice Healthcare for Adults 55+

AI Voice Companion & Clinician Decision Support Portal

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PROJECT ACCESS & LIVE DEMO LINKS

Live Application (Render): <https://carevoice-ai-v54n.onrender.com>

GitHub Repository: <https://github.com/ishitagautam298-droid/carevoice-ai>

1. Problem Statement & Mission

Adults aged 55+ often manage multiple complex chronic conditions (such as hypertension, congestive heart failure, diabetes, and osteoarthritis) while facing barriers with touchscreen navigation and tiny mobile text. Meanwhile, doctors are overwhelmed by unorganized patient communications and raw conversation logs.

CareVoice AI bridges this critical gap by delivering:

1. A warm, empathetic, low-latency conversational voice agent tailored for seniors with zero typing friction.
2. An automated clinical intelligence engine that converts voice conversations into structured, actionable SOAP clinical summaries, quantified longitudinal trends, and multi-tier triage flags without clinician burden.

2. Senior-Centered Voice Experience (55+)

- **Natural Two-Way Speech:** Speech-to-Text (STT) and warm Speech Synthesis (TTS) at a relaxed 0.9x pacing designed for older adults.
- **Senior Accessibility Controls:** 3-level dynamic typography (Standard, Large, XL), WCAG AAA High Contrast mode, speech pace adjustments, and audio replay.
- **Pulsing Audio-Reactive Orb:** Real-time Web Audio API visual feedback reacting to ambient voice levels with zero audio feedback loop.
- **5 Guided Routine Modules:** Daily Wellness Check-in, Medication & Side Effect Log, Symptom Tracker, Doctor Appointment Prep, and Free Conversation.
- **Approved Health Education Library:** Evidence-based AGS/AHA wellness guides (Heart fluid limits, DASH diet, Hydration, Fall safety) with interactive voice read-aloud.
- **1-Touch Emergency SOS:** Immediate direct 911 dispatching and primary caregiver phone dialing in acute safety scenarios.

3. Automated Clinical Triageing & Safety Guardrails

RED FLAGS (URGENT) • Acute Chest Pressure • Severe Orthopnea (3+ pillows) • Syncope / Fall Events • Acute Shortness of Breath Action: Immediate safety triage & emergency protocol	YELLOW FLAGS (MONITOR) • Voluntary Diuretic Omission • Peripheral Ankle Edema • Sleep Fragmentation (<5h) • Postural Lightheadedness Action: Care team callback & dosage review.	POSITIVE PROGRESS • 100% Med Adherence Streak • Daily Walking Goals Met • Stable Dry Weight Log • Improved Energy / Mood Action: Positive reinforcement & adherence maintenance.
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4. Clinician Intelligence & Decision Support Hub

- **Executive Scorecard:** Wellbeing Index (out of 10), 7-day adherence rate %, and active triage status.
- **Longitudinal Trends (7D Charts):** Interactive Recharts for Blood Pressure vs. Weight, Medication Adherence %, Sleep vs. Pain, and Hydration vs. Steps.
- **Structured Patient Agenda:** Captures explicit questions and concerns the patient wants to discuss with their doctor.
- **Actionable Follow-up Checklist:** Evidence-backed clinical action items (lab orders, volume checks, medication reconciliation).
- **EHR SOAP Note Synthesis:** Automated Subjective, Objective, Assessment, and Plan documentation formatted for instant EHR clipboard export.
- **1-Click PDF Encounter Report:** High-resolution downloadable clinical document for patient charts.

5. Multi-Patient Clinical Personas

1. Eleanor Vance (74 y/o): Congestive Heart Failure (Stage B) & Hypertension. Reports orthopnea (3 pillows), +5.5 lbs weight gain, missed Lasix.
2. Arthur Chen (68 y/o): Type 2 Diabetes Mellitus & Osteoarthritis. 95% adherence, preparing for endocrinologist visit, morning knee stiffness.
3. Margaret Miller (81 y/o): Post Total Hip Arthroplasty & Osteoporosis. Mobility milestones, pain medication taper inquiries.

6. Technical Stack & Deployment Pipeline

- Frontend: React 18, TypeScript, Tailwind CSS, Lucide Icons, Framer Motion
- Voice & Audio Engine: Web Speech API (STT & TTS), Web Audio API (zero-feedback analyser)
- Data Visualization & PDF: Recharts, jsPDF
- Backend & Deployment: Express.js (Node.js), render.yaml, Deployed on Render Cloud Platform